

PLANET FITNESS MEMBER RETENTION & ENGAGEMENT ANALYSIS

Data-Driven Insights with Innovative Concepts for Growth

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PROJECT OVERVIEW

This data analytics project explores **member retention and gym usage behavior** at Planet Fitness **using a multi-year synthetic dataset**. The goal was to simulate real-world operational challenges faced by a fitness chain and apply **data-driven insights to improve member engagement**, reduce churn, and optimize resources.

Key components of the project include **exploratory analysis, predictive modeling, and behavioral segmentation**.



DATA & METHODOLOGY

Data simulated:

- 1,500 members, 100,000+ check-ins

Tools:

- Python, Pandas, scikit-learn, Seaborn

Method overview:

- Retention analysis
- Churn prediction
- Usage heatmaps
- Scenario modeling

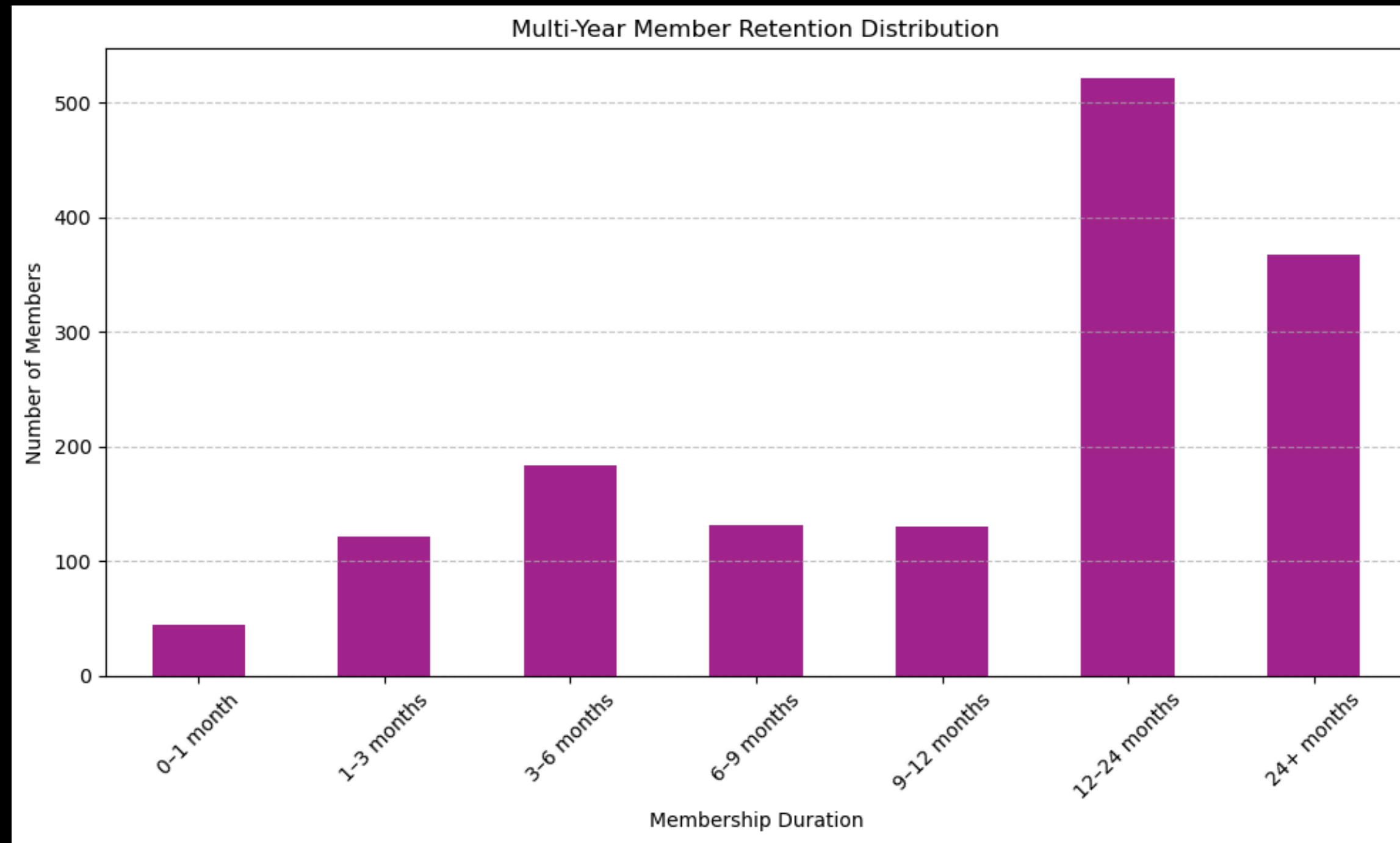


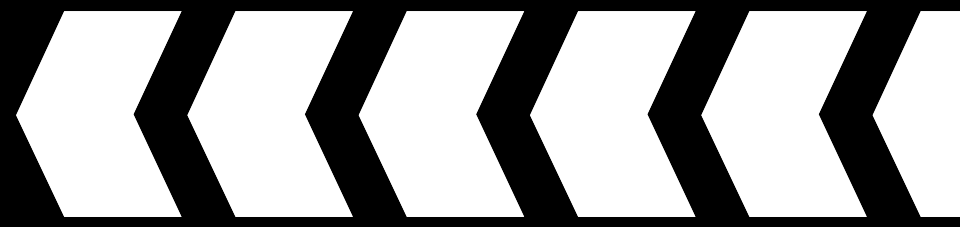
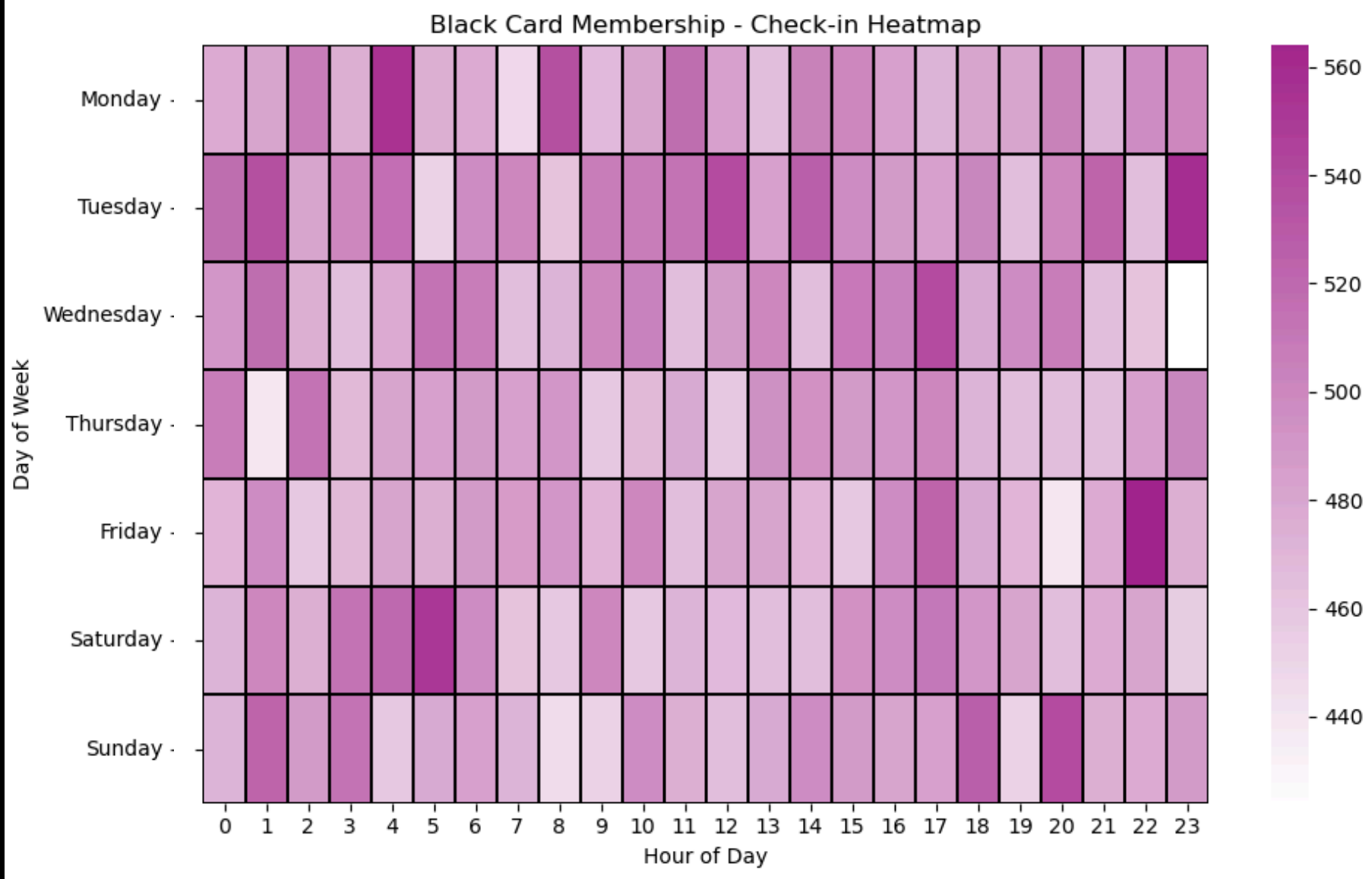
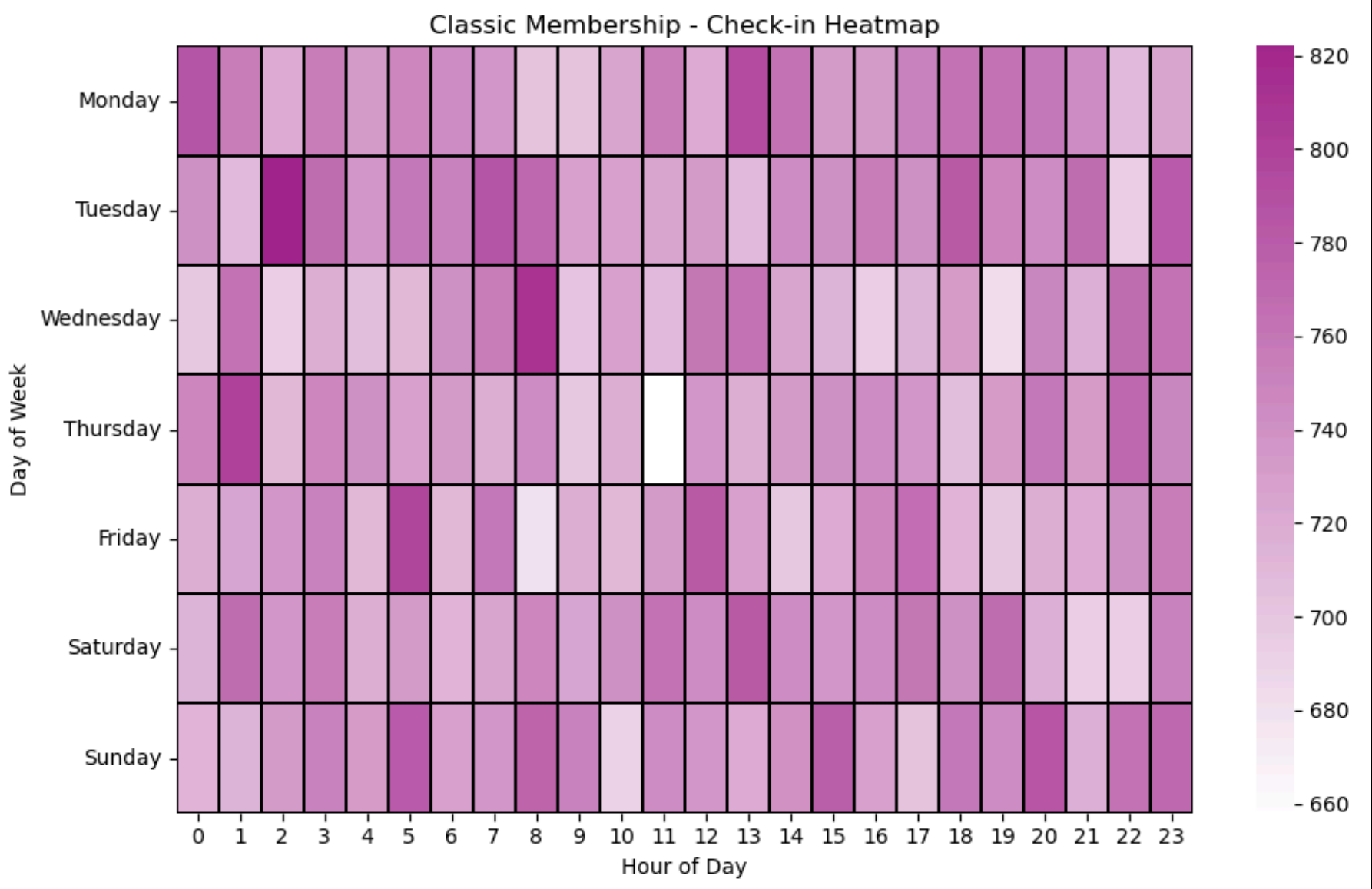
KEY FINDINGS

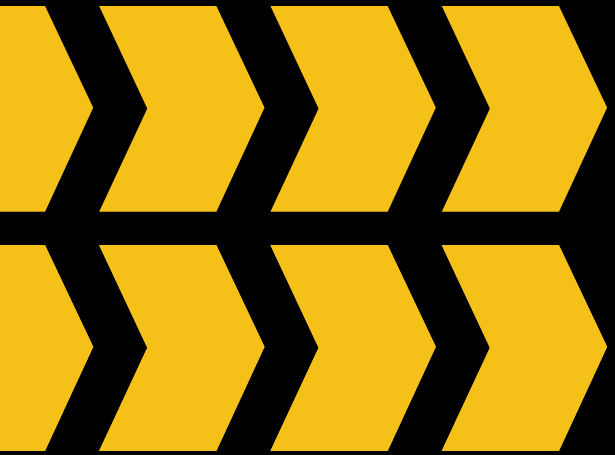
- Retention insights (e.g., **first 30 days are critical**)
- Churn model performance (**83% accuracy**)
- **Usage patterns** by hour, day, and membership type
- Visuals: retention curve, confusion matrix, **heatmap**



VISUALIZATIONS







STRATEGIC RECOMMENDATIONS

Target At-Risk Members Early

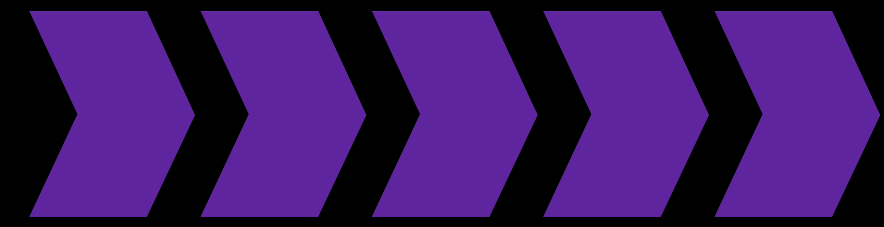
- Use churn prediction models to identify members with low check-in frequency in their first 30 days. Target them with automated follow-ups, check-in challenges, or support.

Optimize Staffing by Usage Patterns

- Heatmap analysis reveals consistent daily and hourly trends (e.g., 6–9 AM and 5–8 PM peaks). Use this data to optimize staffing, cleaning schedules, and maintenance planning.

Segment Strategies by Membership Type

- Classic and Black Card members show different usage behaviors. Tailor retention campaigns and incentives based on usage frequency and time-of-day preferences.



CONTACT

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The image is a composite of two photographs. On the left, a woman with brown hair tied back, wearing a black sports bra and leggings, is smiling while holding a large black weight plate. She is in a gym setting with metal racks visible in the background. On the right, a woman with dark hair in a bun, wearing a black sports bra, is performing a burpee on a paved surface. A large black diagonal bar runs across the center of the image, serving as a background for the text.

THANK YOU

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